

Effects of surface characteristics on infiltration patterns in an arid shrub desert

Xin-Ping Wang,^{1*} Xin-Rong Li,¹ Hong-Lang Xiao,¹ Ronny Berndtsson² and Yan-Xia Pan¹

¹ Shapotou Desert Experimental Research Station, Cold and Arid Regions Environmental and Engineering Research Institute, Chinese Academy of Sciences, 260 Donggang West Road, Lanzhou 730000, People's Republic of China

² Department of Water Resources Engineering, Lund University, S-22100 Lund, Sweden

Abstract:

Precipitation is often the sole source of water replenishment in arid and semi-arid areas and, thus, plays a pertinent role in sustaining desert ecosystems. Revegetation over 40 years using mainly *Artemisia ordosica* and *Caragana korshinskii* at Shapotou Desert Experimental Research Station near Lanzhou, China, has established a dwarf-shrub and microbiotic soil crust cover on the stabilized sand dunes. The redistribution of infiltrated moisture through percolation, root extraction, and evapotranspiration pathways was investigated. Three sets of time-domain reflectometry (TDR) probes were inserted horizontally at 5, 10, 15, 20, 30 and 40 cm depths below the ground surface in a soil pit. The three sets of TDR probes were installed in dwarf-shrub sites of *A. ordosica* and *C. korshinskii* community with and without a microbiotic soil crust cover, and an additional set was placed in a bare sand dune area that had neither vegetation nor a microbiotic soil crust present. Volumetric soil moisture content was recorded at hourly intervals and used in the assessment of infiltration for the different surface covers. Infiltration varied greatly, from 7.5 cm to more than 45 cm, depending upon rainfall quantity and soil surface conditions. In the shrub community area without microbiotic soil crust cover, infiltration increased due to preferential flow associated with root tunnels. The microbiotic soil crust cover had a significant negative influence on the infiltration for small rainfall events (~10 mm), restricting the infiltration depth to less than 20 cm and increasing soil moisture content just beneath the soil profile of 10 cm, whereas it was not as strong or clear for larger rainfall events (~60 mm). For small rainfall events, the wetting front depth for the three kinds of surface cover was as follows: shrub community without microbiotic soil crust > bare area > shrub community with microbiotic soil crust. In contrast, for large rainfall events, infiltration was similar in shrub communities with and without microbiotic soil crust cover, but significantly higher than measured in the bare area. Soil water extraction by roots associated with evapotranspiration restricted the wetting front penetration after 1 to 3 h of rainfall. Copyright © 2006 John Wiley & Sons, Ltd.

KEY WORDS desert ecosystem; soil moisture; infiltration; revegetation; microbiotic soil crust; dwarf-shrub-stabilized sand dune; Shapotou; Tengger Desert

Received 16 November 2004; Accepted 15 August 2005

INTRODUCTION

The hydrologic cycle is the main plant community shaping process in arid ecosystems. The infiltration process, especially, forms the basis for transferring limited rainfall amounts to the soil system available for plant roots. The importance of understanding infiltration patterns and processes cannot be overstated considering its relevance to soil water availability, where precipitation is often the sole source of water replenishment. Owing to large spatial variability of soil characteristics and surface cover, experimental methods are needed to measure infiltration at several locations simultaneously during rainfall and/or irrigation. Studies in arid northwest China have shown that a change in landscape from migrating desert dunes to artificially stabilized shrub-covered dunes will almost always result in a reduction of water yield, especially in soil water replenishment, where groundwater does not support the natural vegetation cover (Wang *et al.*,

2004). Therefore, it is especially important to be able to quantify changes in infiltration properties by a reliable observation and measurement method for these areas. Previous research has indicated the feasibility of using time-domain reflectometry (TDR) probes installed vertically into the soil to track the propagation of the wetting front during infiltration (Topp *et al.*, 1982; Noborio *et al.*, 1996; Timlin and Pachepsky, 2002). They used TDR probes installed vertically into the soil to track the propagation of the wetting front during infiltration. A rapid and approximate prediction of the wetting front advance, and thus the time for the wetting front to reach a certain depth, is often needed to model water infiltration in soils (Moldrup *et al.*, 1996).

Previous studies have shown that subsurface heterogeneities, such as worm holes and macropores formed by decaying root matter, can contribute to preferential flow pathways that decrease the time for infiltrating water to reach the water table (Geiger and Durnford, 2000; Devitt and Smith, 2002). However, infiltration patterns between vegetation with the presence and absence of microbiotic soil crust cover have not yet been considered. The soil moisture field exhibits substantial temporal variability

* Correspondence to: Xin-Ping Wang, Shapotou Desert Experimental Research Station, Cold & Arid Regions Environmental and Engineering Research Institute, CAS, 260 West Donggang Rd, Lanzhou 730000, PR China. E-mail: xpwang@lzb.ac.cn

in arid and semi-arid environments, and soil moisture is the most critical factor controlling plant productivity and reproduction in arid to semi-arid environments (Noy-Meir, 1973; Rodriguez-Iturbe, 2000). Therefore, single surveys or repeat sampling at intervals unrelated to wetting and drying cycles provides limited information about soil moisture dynamics (Schlesinger *et al.*, 1990; Kieft *et al.*, 1998).

Water movement is a critical process in semi-arid and arid landscapes characterized by a patchy distribution of essential resources (Schlesinger *et al.*, 1990). Given the importance of water flow in the formation and maintenance of patchy landscapes, it follows that the factors influencing the amount and distribution of infiltration ultimately affect the pattern phenomenon itself. As water availability and landscape patchiness are important components to these landscapes, a proper evaluation of the relationships between rainfall, infiltration, and surface cover is crucial to understanding how water is distributed throughout the landscape. Infiltration is influenced by soil surface characteristics (such as microtopography, cracking, surface sealing and crusting), soil physical properties (such as bulk density, organic matter and particle size distribution), and the characteristics of the vegetation (such as plant biomass and cover) (Eldridge *et al.*, 2000).

In this study, we compared infiltration and soil moisture content between shrub community sites with and without microbiotic crust covers and the bare surface area of a sand dune. This enabled us to characterize the soil moisture state immediately following rainfall events in shrub land (with and without microbiotic soil crust cover) and bare area (interspace of shrub land), so that we could assess how the changes of surface cover influence the spatial distribution of soil water availability. We defined 'infiltration' as the total equivalent depth of water that enters the soil (units of length), resulting either from a single rainfall event or a series of events. Our goal was to understand how an artificially stabilized sand dune with shrub communities affects the amount of soil water that is available to plants. The first section of the paper thus describes the experiments and methods used. Then follows results based on experimental observations. Finally, we conclude the paper with a discussion of practical results.

STUDY AREA

The study was conducted at Shapotou in Ningxia Hui Autonomous Region at the southeast edge of the Tengger Desert at 37°32'N and 105°02'E with an elevation of 1339 m a.m.s.l., belonging to the steppe desert zone, an area that also can be described as an ecotone between desert and oasis (Li *et al.*, 2002; Shapotou Desert Experimental Research Station, Chinese Academy of Sciences, 1991). Based on long-term measurements from 1955 to 2001, the mean temperature in January is -6.1°C and in July 24.7°C . Mean relative humidity varies from 31% (April) to 54% (August). The first

frost occurs in late September and the last frost ends in mid April. The annual mean precipitation is 191 mm and about 80% of this falls between June and September. The annual class A pan evaporation is about 2400 mm. The area has large and dense reticulate dune chains. The main dune crest migrates southeastward at a velocity of $0.3\text{--}0.6\text{ m year}^{-1}$. The soil is loose and impoverished mobile sand with a consistent moisture content ranging from 3 to 4% and classified as Typic Psammaquents (Berndtsson *et al.*, 1996). The depth to groundwater is between 50 and 80 m and, therefore, is unable to support the natural vegetation. Rainfall is the only source of fresh water. The natural predominant plants are *Hedysarum scoparium* and *Agriophyllum squarrosum* with a cover of approximately 1–2% (Shapotou Desert Experimental Research Station, Chinese Academy of Sciences, 1991).

Revegetation procedure

To ensure unimpeded passage of the east–west Baotou–Lanzhou railway through the sand dunes in the Shapotou region, a 16 km long vegetated protection system was established in the 1950s, with widths of 500 m to the north side and 200 m to the south side of the railway. Initially, a sand barrier was established with woven willow branches or bamboo to reduce wind erosion. Behind the sand barrier, straw chequer-boards (wheat or rice straw) were installed, usually with sections of 1 m^2 in area. The straw structures remain intact for 4–5 years, facilitating the establishment of planted xerophytic shrubs typical in this highly eroded environment. Straw chequer-boards at a height of 0.15–0.2 m above the ground increase the roughness of the sand surface by 400–600 times, and they reduce the wind velocity by 20–40% at a height of 0.5 m and by 10% at a height of 2 m above the surface (Zou *et al.*, 1981). The quantity of sand transported over a straw chequer-board is only 1% of that over uncovered mobile sand dunes (Zhu *et al.*, 1992). Soil crust containing mosses can resist erosion forces from wind at speeds as high as 25 m s^{-1} (Shapotou Desert Experimental Research Station, Chinese Academy of Sciences, 1991). The fixed sand surface provides better conditions for the recolonization by many annual plants and has led to the formation of microbiotic soil crusts. The colonization and spread of a large number of species have turned the formerly moving dune-dominated landscape into a complex reconstructed desert ecosystem (Li *et al.*, 2002). The formation and effects of the soil crust phenomenon have attracted the concern of many researchers during recent years owing to its importance and its significant effect on hydrology and land use (e.g. Li *et al.*, 2002). The shrubs *Artemisia ordosica*, *Caragana korshinskii*, and *H. scoparium* were planted in the centre of the straw chequer-board sections and grown without irrigation. Hence, three main kinds of shrub land (without microbiotic soil crust cover (Figure 1a) or with microbiotic soil crust cover (Figure 1b), and bare area (Figure 1c) (interspace of shrub land) were formed and these three surface types are investigated in this paper.

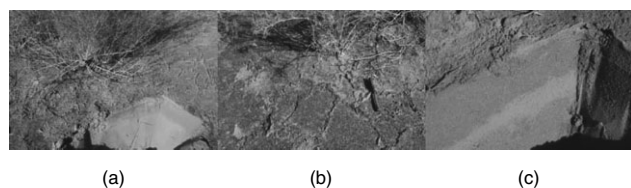


Figure 1. The three main kinds of land cover around the Shapotou area: (a) shrub presence without crust cover; (b) shrub presence with crust cover; (c) neither shrub presence nor crust cover

EXPERIMENTAL PROCEDURE

Data collection

At each experimental site with different surface cover type, TDR coated burial probes were inserted (Model 6050X1, Trase System I, Soilmoisture Equipment Corp., USA) at six depths: 5, 10, 15, 20, 30, and 40 cm. The probe has three 20 cm rods spaced 2.5 cm apart. For probe installation, a pit was dug in the soil of sufficient width to be able to insert the probes. The probes were inserted into the soil through the side of the pit that was unaltered and were placed horizontally in the direction of maximum slope of the terrain. Once the probes had been inserted, the pit was carefully refilled, avoiding perturbations as far as possible, and the surface was contoured similar to the surrounding slope. The sites were set up in September 2002 and measurements were not begun until 6 months later with a view to allowing the soil to settle to approximately the original soil density. We used the six probes at each site to estimate the average soil moisture content throughout the top 45 cm of the soil. The value from each probe was weighted assuming it represents soil moisture over depth intervals of 0–7.5, 7.5–12.5, 12.5–17.5, 17.5–25, 25–35, and 35–45 cm, a reasonable assumption if soil wetting is uniform. These depth intervals were chosen according to the endpoints between each probe and to place equal weighing at each depth.

Rainfall R may be separated into the components runoff, infiltration I , interception (rainfall that is caught on the shrub surfaces), and evapotranspiration (ET). The ET is defined as the sum of transpiration and evaporation from plant and soil surfaces. Interception and runoff are often disregarded in this specific arid environment because they represent a small portion of the total rainfall. In the present condition, no runoff occurs on the sand dune surface, the interception is around 6.9% and 11.7% of the simultaneous overall precipitation (Wang *et al.*, 2005), and the groundwater is deep, i.e. there is no evaporation from water table. These simplifications lead to the approximation that rainfall is nearly equal to the amount of infiltration plus ET, $R \cong I + ET$.

Effective rainfall is defined here as the amount of rainfall stored in the shrub root zone. Rainfall that runs off the soil surface or passes through the root zone does not contribute to shrub growth and biomass, and neither does water that evaporates. There are two steps to estimating effective rainfall. First, the total rainfall amount must be measured accurately. Second,

the amount stored in the root zone must be determined. Both infiltration and rainfall can be measured. Therefore, ET during the infiltration processes can be calculated by the equation mentioned above. Taking into account the fact that soil moisture storage is a transient process, it is reasonable to set that the amount of moisture in the root zone before the rainfall has an influence on the balance between rainfall that is stored in the root zone and the amount that percolates through it. Infiltrated water can either refill the soil profile or, if the profile cannot hold the infiltrated water, the excess percolates below the root zone. If the infiltrated depth is greater than what the root zone can hold, then the effective rainfall equals the amount the root zone can hold when the soil profile is fully replenished. If the amount the root zone can hold is greater than the amount that infiltrates, then the effective rainfall equals the infiltrated depth and the available water is increased by that amount.

We measured infiltration during and following each rainfall event by combining water content and wetting-front measurements. The period of observation was 1 April to 31 August 2003, which included most of the annual rainfall. Daily rainfall was recorded with an autorecording siphon rain-gauge linked to a data logger; the values were summed every 1 h for analysis. The error of the siphon rain-gauge was ± 0.2 mm for rainfall < 10 mm and ± 0.02 mm for rainfall ≥ 10 mm. The wetting-front location and cumulative infiltration were detected by measuring changes in water content in the soil profile (e.g. Noborio *et al.*, 1996). Noborio *et al.* (1996) found that the cumulative infiltration estimated by TDR compared favourably with observed infiltration, and the distance to the wetting front from the soil surface estimated by TDR agreed well with the observed values. All data were collected simultaneously during and directly following rain events sufficient to trace wetting-front changes on a 5 cm scale. The wetting depth was measured at the same locations as water content. The wetting front was determined by evaluating the difference in the soil moisture content at the different corresponding depth layers. Root samples were taken by using a special root sampling equipment with 10 cm diameter and 10 cm depth in three radially aligned directions (i.e. in an equal angle of 120°). The sampling locations in each direction were set from the point close to the plant stem to 100 cm apart with an intervals of 20, 50, and 100 cm horizontally to a depth of 150 cm with an interval of 10 cm vertically each. Root length is the sum of the single root at specific layers measured directly with a ruler. The data of root weight were processed by weighing method; the weight proportions to the area of each annulus were 0.04, 0.21, and 0.75 for the innermost, middle, and farthest samplers and consistent with the collecting area they represent. Root weight is the sum of single roots at specific layers after drying for 8 h at 60°C in an oven.

For the three different surface covers, the infiltration I , or the total equivalent water depth that enters the soil following a rainfall event, was calculated using Bhark

and Small's (2003) formulation according to

$$I = \int_{z=0}^{S_f} (\theta_t(z) - \theta_i(z)) dz \quad (1)$$

where θ_i is the initial volumetric soil moisture before the rainfall event, θ_t is volumetric soil moisture at some time t after the rainfall event, S_f (cm) is the wetting-front depth, and z is depth below the soil surface. Bhark and Small (2003) approximated I based on the Green and Ampt (1911) model:

$$I = \int_{z=0}^{S_f} (\theta_t(z) - \theta_i(z)) dz \approx (\theta_t - \theta_i) S_f \quad (2)$$

In the current experiment, the corresponding wetting-front depth D for layers at 5 cm, 10 cm, 15 cm, 20 cm, 30 cm, and 40 cm was set at 7.5 cm, 5 cm, 5 cm, 7.5 cm, 10 cm, and 10 cm respectively. So, the cumulative I during individual rainfall events could be described as

$$I = \sum (\theta_{D_{\max}} - \theta_{D_i}) D \quad (D = 7.5, 5, 5, 7.5, 10, 10 \text{ cm}) \quad (3)$$

where $\theta_{D_{\max}}$ is the maximum volumetric soil moisture content at the soil layer D after the rainfall event and θ_{D_i} is the volumetric soil moisture at the soil layer D before the rainfall event. The maximum value following each event was used for our infiltration calculations. We consequently compared infiltration from shallow soil profiles in the three different surface cover types.

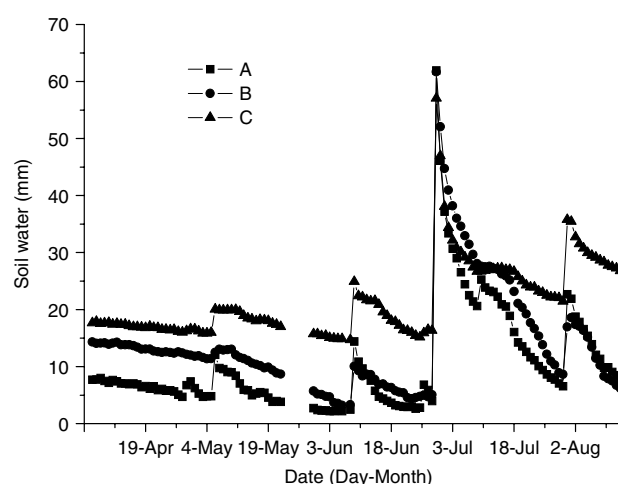


Figure 2. Comparison of total water content time series in the 0–45 cm soil horizon depending on surface cover (A: shrub presence without crust cover; B: shrub presence with crust cover; C: neither shrub presence nor crust cover)

RESULTS AND DISCUSSION

Four rainfall events were observed during the observation period (1 April to 31 August 2003), with totals of 6.0 mm, 10.3 mm, 4.0 mm, and 57.5 mm (Table I). The respective average rainfall intensities were 0.6 mm h⁻¹, 3.4 mm h⁻¹, 0.5 mm h⁻¹, and 6.4 mm h⁻¹. The observed rainfall events can be said to represent typical monsoon convective rainfall occurring in the area (Wang *et al.*, 2001). Figure 2 shows a general overview of the total water content time-series in the 0–45 cm soil horizon depending on surface cover

Table I. Changes in soil water content (mm) in the different soil layers as affected by rainfall at the various sites

	5–6 May	9 June	25 June	28–29 June	Total
Rainfall (mm)	6.0	10.3	4.0	57.5	77.8
<i>A: shrub presence without crust cover</i>					
Layer 0–7.5 cm	5.4	6.3	3.1	14	28.8
Layer 7.5–12.5 cm	0.2	0.6	0	7	7.8
Layer 12.5–17.5 cm	0	0.1	0	5.1	5.2
Layer 17.5–25 cm	0	0	0	6.2	6.2
Layer 25–35 cm	0	0	0	11.8	11.8
Layer 35–45 cm	0	0	0	12	12
Total change in 45 cm (mm)	5.6	7	3.1	56.1	71.8
<i>B: shrub presence with crust cover</i>					
Layer 0–7.5 cm	1.9	6	0.2	11	19.1
Layer 7.5–12.5 cm	0.1	0.2	0	7.9	8.2
Layer 12.5–17.5 cm	0	0	0	7	7
Layer 17.5–25 cm	0	0	0	7	7
Layer 25–35 cm	0	0	0	11.7	11.7
Layer 35–45 cm	0	0	0	12.3	12.3
Total change in 45 cm (mm)	2	6.2	0.2	56.9	65.3
<i>C: neither shrub presence nor crust cover</i>					
Layer 0–7.5 cm	4.2	8.2	0.8	12.4	25.6
Layer 7.5–12.5 cm	0.1	0.2	0	7.2	7.5
Layer 12.5–17.5 cm	0	0.1	0	6.2	6.3
Layer 17.5–25 cm	0	0	0	5.2	5.2
Layer 25–35 cm	0	0	0	7.7	7.7
Layer 35–45 cm	0	0	0	6.9	6.9
Total change in 45 cm (mm)	4.3	8.5	0.8	45.6	59.2

(A: shrub community without microbiotic surface crust; B: shrub community with crust cover; C: bare surface). As seen from Figure 2, the bare dune surface area has the largest soil water content throughout the entire observation period, though the initial soil water content was different for profiles A, B, and C. Increases in peak soil water content just after rainfall are less accentuated for the area with surface crust (B) due to immediate evapotranspiration by crust components. This is mainly observed for smaller rainfall events. Therefore, the effect of the crust in time is to reduce the total soil water content compared with the shrub area without crust. A more detailed analysis of effects of the different soil surface cover types is given below.

Difference in soil physical properties

A soil composition analysis of the stabilized dune sand without and with microbiotic soil crust (A, B) and the stabilized bare dune sand (C) (Figure 3) showed that the particle size distribution, especially for soil profiles 0–5 and 5–10 cm, differs significantly between the surface types. At a soil depth of 0–5 cm, the site with microbiotic soil crust contained the largest amount of fine soil particles in the range of 0.002–0.25 mm. The largest difference occurs for fine soil particles in the range 0.002–0.1 mm. At a soil depth of 5–10 cm, the difference between A and B was not as pronounced, and at a soil depth of 10–20 cm there was no difference between A, B, and C.

The organic matter content and bulk density displayed significant differences between the different surface types. Differences could be noted down to the 20–30 cm soil depth (Figure 4). As expected, the organic matter content was highest for the site with microbiotic surface crust (B), followed by the site without microbiotic surface crust (A). Bulk density decreased significantly after dune stabilization, as seen from Figure 4a, and porosity increased by 10 to 20% (data not shown). This suggested that the soil physical properties play a

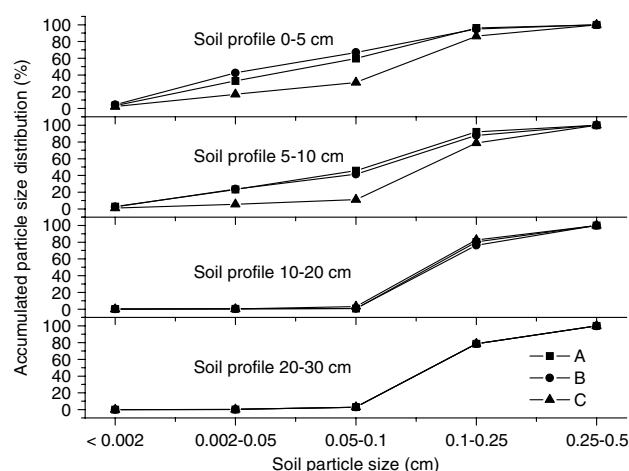


Figure 3. Comparison of the accumulated particle size distribution from sites of stabilized dune sand (A: shrub presence without crust cover; B: shrub presence with crust cover; C: neither shrub presence nor crust cover)

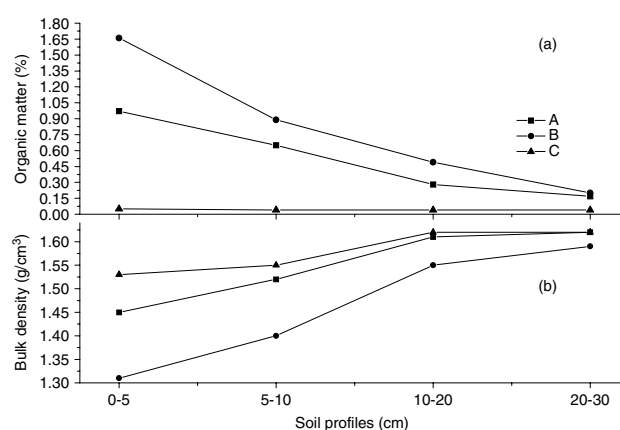


Figure 4. Comparison of (a) organic matter and (b) bulk density from sites of stabilized dune sand (A: shrub presence without crust cover; B: shrub presence with crust cover; C: neither shrub presence nor crust cover)

significant role in affecting the water-holding capacity and, subsequently, the speed of the wetting-front advance. Our results corroborate the findings of Eldridge (1993), who suggested that, in the semi-arid woodlands of eastern Australia, differences in the hydrological responses to cryptogam covers are attributed to differences in soil physical properties, particularly porosity and aggregate stability, and the different pathways by which infiltrating water accesses the soil profile.

Change in soil water content

Studies by Abu-Awwad (1997) indicated that soils under different surface covers are influenced by the water-holding capacity as affected by the soil surface crust. Also, the potential ET is very high during the summer season. Therefore, changes in soil water content with time are considered of great interest. The change in soil water content with time during April to August 2003 at the three sites A (shrubs community without microbiotic soil crust), B (shrubs community with microbiotic soil crust) and C (bare area) are shown in Table I for soil layers 0–7.5, 7.5–12.5, 12.5–17.5, 17.5–25, 25–35, and 35–45 cm. Soil water content changes at the three sites followed a similar trend. Soil moisture content rose to a maximum typically within about 1 h after each rainfall event, followed by a slow decline due to vertical redistribution and ET. Only after one major rainfall (28–29 June, 57.5 mm) did the soil water content change below 45 cm.

In Table I, the rainfall amount is compared with the total change of soil water content in the 0–45 cm soil section for the different soil surface categories. As seen from the table, the rainfall is always greater than the total soil water content, the difference thus representing the ET. This difference (ET) is always greatest for the area with soil surface crust, except for the very large rainfall event (28–29 June). Consequently, it can be said that the surface crust induces a greater ET for small- and medium-sized rainfall events. This is logical, since the surface crust prevents infiltration at the soil surface and

also the biological components here are adding up to the total transpiration rate for the entire soil profile.

Infiltration patterns

Figure 5 shows soil water content profiles for the major rainfall event (57.5 mm) occurring during the observation period. As seen from the figure, there is a marked difference in soil water content for soils deeper than about 15 cm during dry periods (i.e. there is no rainfall for 330 h). The stabilized bare dune sand has a higher soil water content (up to about 5%), whereas the plant roots of the shrub-covered sites effectively utilize the excess water at these depths. During dry periods there is also a marked difference between the shrub sites. The site without microbiotic crust cover usually has two to three times the soil water content (an absolute difference of about 2%) compared with the crust-covered site at 0–10 cm depth. This indicates the water uptake and water use by the crust.

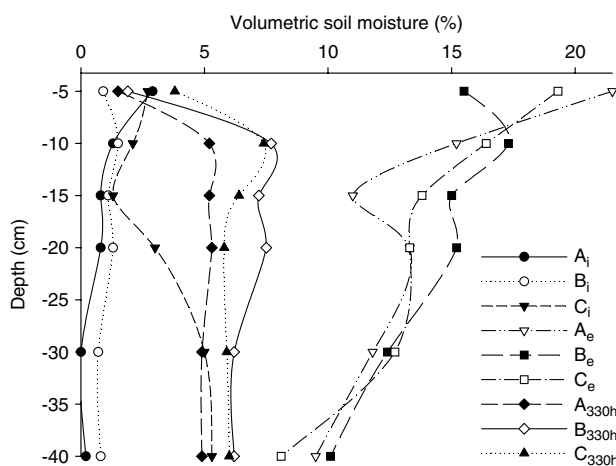


Figure 5. The soil moisture profile (5–40 cm) of sites A, B, and C at the initial stage (A_i , B_i , C_i), the end stage of the cessation of rainfall (A_e , B_e , C_e), and 330 h (A_{330h} , B_{330h} , C_{330h}) after the cessation of rainfall ($P = 57.5$ mm)

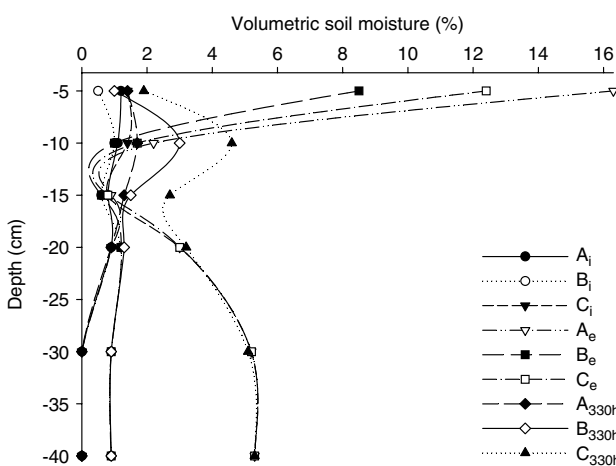


Figure 6. The soil moisture profile (5–40 cm) of sites A, B, and C at the initial stage (A_i , B_i , C_i), the end stage of the cessation of rainfall (A_e , B_e , C_e), and 330 h (A_{330h} , B_{330h} , C_{330h}) after the cessation of rainfall ($P = 10.3$ mm)

Under a relatively higher magnitude of precipitation (57.5 mm), the shrub site community with microbiotic soil crust cover has a relatively higher soil moisture content profile below 10 cm at the end of rainfall and after the rainfall cessation for 330 h (Figure 5), except for the top soil layer of 5 cm. This suggests that the microbiotic soil crust may contribute to a significant infiltration by reducing the evaporation that occurs from the shallow soil layer of less than 10 cm, compared with sites without microbiotic crust cover.

For smaller rainfall events (10 mm), the soil moisture of stabilized bare dune sand below 30 cm is maintained at around 5.3% (Figure 6), suggesting that the dry and wet cycling regimes for this kind of surface cover may not exceed a depth of 30 cm. Plant roots for the shrub sites, however, will tend to decrease the soil water content at depths greater than 30 cm.

Infiltration controlled by the presence or absence of vegetation

The change of soil moisture content of the entire soil profile down to 45 cm was 1.2 times greater at sites A and B than at site C throughout all individual rainfall events, until 330 h after cessation of rainfall (Figure 5), though the soil moisture content at the top soil layer 0–5 cm of site B was lower than that of sites A and C. Here, the top soil layer consistently displayed lower soil water contents, indicating water uptake and ET by the crust.

Figure 7 shows the root length and weight dependence on depth. As seen from the figure, most of the plant roots were concentrated in the layer between 0 and 60 cm below the surface. Both shrub types (*A. ordosica* and *C. korshinskii*) account for about 88% of the total root amount down to 60 cm. Consequently, root tunnel bypass flow and root water extraction can affect infiltration and redistribution processes within and after rainfall cessation (Figure 5). So, the presence of vegetation in addition to the microbiotic soil crust may contribute to the higher soil moisture beneath the soil surface both during and after the rainfall events at the relatively higher overall precipitation and rainfall intensity of 57.5 mm and 6.4 mm h^{-1} respectively. This strongly supports previous studies showing that the water transmission resulting from increased activity far exceeds water retention when microbial bodies were swelled by rainwater (Belnap and Gardner, 1993; Williams *et al.*, 1999).

The presence of a microbiotic soil crust, together with the shrub communities, improves the physical surface structure of the sand dunes stabilized by (1) decreasing soil particle size, (2) decreasing bulk density, (3) increasing organic matter content, and (4) increasing volumetric soil moisture in the near-surface environment, thus contributing to differences in infiltration patterns during rainfall events and soil water redistribution at the cessation of rainfall. This was seen to lead to a larger soil moisture content beneath 10 cm for areas with microbiotic crust (Figure 5). Subsequently, annual herbs such as *Eragrostis poaeoides*, *Bassia dasyphylla*, *Corispermum patelliforme*, *Corispermum*

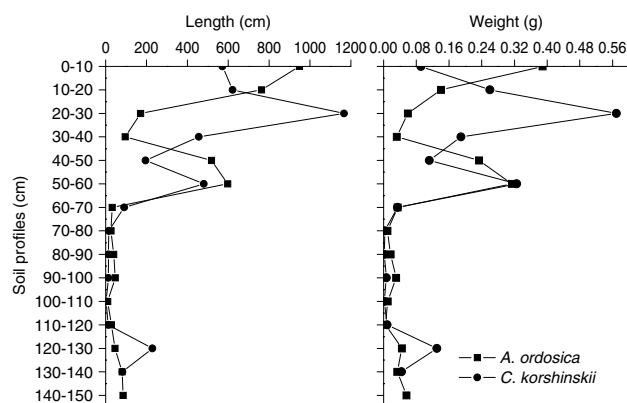


Figure 7. Root distribution (the overall root length and weight within sampling soil column 100 cm apart from the plant stem) of shrub plants *A. ordosica* and *C. korshinskii* in the soil column with 10 cm diameter

mongolicum, *A. squarrosus*, *Setaria viridis*, and the seedlings of the semi-shrub *A. ordosica* were seen to emerge in these areas (sites A and B) after individual rainfall events. In contrast, there were no seedlings emerging or growing in the stabilized bare dune area (site C).

SUMMARY

Based on a technique using straw chequer-boards and planting xerophytic shrubs to revegetate areas in the southeastern edge of the Tengger Desert, we conclude that the method in which sand dunes are stabilized significantly affects the soil hydrological properties. Stabilization and revegetation accelerates environmental improvement, particularly affecting the upper layer soil formation processes, such as microbiotic soil crusts. Meanwhile, the establishment of a microbiotic soil crust enhances the stability of surface sand (Li *et al.*, 2003). The results showed that there were marked differences in infiltration patterns in response to changes in shrubs and microbiotic soil crust present at three sites with varying soil surface conditions. The results indicate that the differences in response to the shrubs and microbiotic soil crust present are probably due to differences in soil physical properties, such as organic carbon, where the presence of organic matter (loam) and vegetation will affect infiltration and runoff (Dunkerley, 2002), generally increasing infiltration. The particle size distribution in the upper soil layers, associated with the three soil surface conditions, also influenced infiltration. Taken together, the results indicate that, as the soil surface condition and soil physical conditions improve from those associated with shifting sand dunes, the microbiotic soil crust cover, like the dwarf-shrub cover, becomes increasingly important as a descriptor of soil hydrological properties. These new insights on how microbiotic soil crusts affect infiltration in soils can contribute to a better understanding of the hydrological response at the plot scale and to an improvement of the calculation of plant-available soil water.

ACKNOWLEDGEMENTS

The research projects on which this paper is based were funded by the National Natural Science Foundation of China (grant nos. 40301008 and 90202015), the Innovation Research Foundation of Chinese Academy of Sciences (grant no. KZCX3-SW-324), and the Talent Training Program for Young Scientist in West China of Chinese Academy of Sciences. R. Berndtsson was partly funded by Swedish Science Research Council. We are indebted to Professor David Mouat and two other anonymous referees for their helpful and constructive comments and suggestions for improving the manuscript.

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